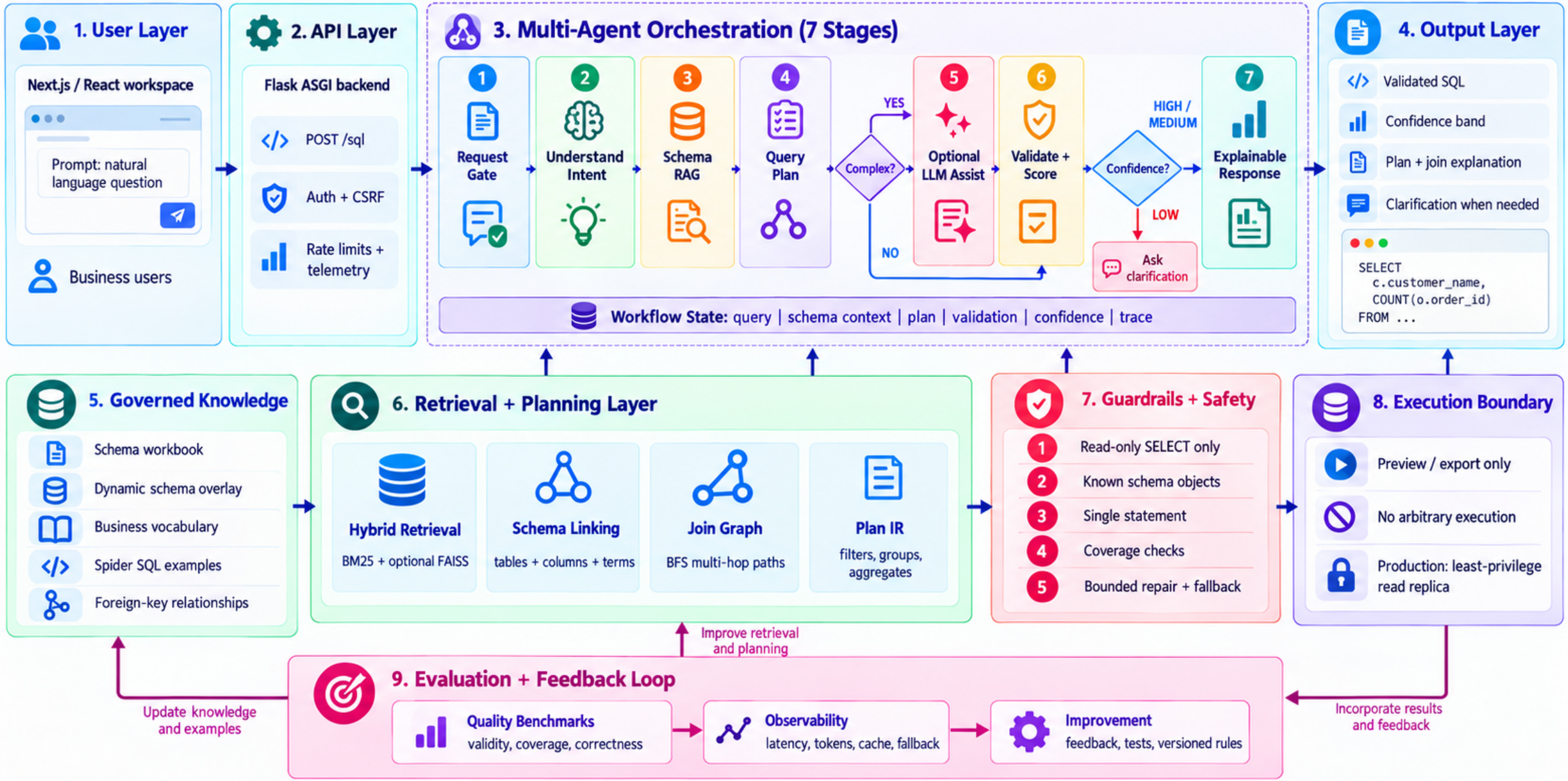
Enterprise SQL Copilot - System Architecture

Planner-first, schema-grounded, validated text-to-SQL



SQL Copilot
Decision-ready analytics

Governed, read-only analytics

Ask with clarity.
Move with confidence.

A focused workspace for exploring your business data, reviewing trusted SQL, and understanding the path behind every answer.

Create your account

Create an account and a recovery check.

Name

Email

Password

Password strength 0%

Confirm password

Account recovery

Security question

What city were you born in?

Your answer

Create account

Already have an account? Sign in

Dashboard
SQL Copilot
SQL Repair
Schema Explorer
Schema Graph
Data Model Studio
Query Execution
Query Planner
SQL Optimizer
Settings

Dashboard

Review request quality, confidence, and recent SQL activity.

29 events | 96.55% success

Relationships | Copilot | Refresh

Planner Accuracy	SQL Accuracy	Validator Precision	Confidence Reliability
46%	96.55%	100%	94.12%
29 evaluated plans	28/29 validated	Read-only policy and schema checks	Outcome-aligned confidence
Details	Details	Details	Details

Last 7 days | Day | Week | Month | Quarter | Year | All

29 events

Query Analytics

Confidence, validation, fallback, and latency over the selected range.

96% valid | 1421ms avg



Recent activity

96.55% success

Headcount by department — Valid — 100% confidence — 0ms

Headcount by department — Valid — 100% confidence — 0ms

Headcount by department — Valid — 100% confidence — 0ms

Headcount by department — Valid — 100% confidence — 0ms

Headcount by department — Valid — 100% confidence — 0ms

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Schema Explorer

Explore tables, columns, and relationships surfaced by the backend schema graph.

Search tables or columns

All domains

Graph view

associate_activity

live

Business delivery activity logged by associates against projects.

COLUMNS

activity_id PK | associate_id FK | project_id FK | activity_date | hours_logged | activity_type | outcome

INDEXES

activity_id | associate_id | project_id | activity_date

RELATIONSHIPS

associate_activity.associate_id -> associates.associate_id

associate_activity.project_id -> projects.project_id

associate_assignments

live

Associate-to-project assignments, roles, allocation, and assignment status.

COLUMNS

assignment_id PK | associate_id FK | project_id FK | assignment_role | allocation_percent | assignment_status | assigned_at | release_date

INDEXES

assignment_id | associate_id | project_id | assignment_status

RELATIONSHIPS

associate_assignments.associate_id -> associates.associate_id

associate_assignments.project_id -> projects.project_id

associate_activity explorer

COLUMNS

activity_id	INTEGER
associate_id	INTEGER
project_id	INTEGER
activity_date	DATE
hours_logged	DECIMAL(8,2)

JOIN RECOMMENDATIONS

associate_activity -> associates

associate_activity -> projects

Inspect in relationship graph

associate_availability

live

Planned associate capacity and availability periods for staffing decisions.

COLUMNS

associates

live

Business associate directory with roles, reporting lines, and employment status.

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Schema Graph

Interactive relationship graph with zoom, pan, minimap, and animated join edges.

Interactive Relationship Graph

16 tables | Auto layout | Highlight join paths



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SQL Copilot Chat

Ask business questions, receive reviewed SQL, and inspect the decision trail.

Conversation

Search chats

Headcount by department

Reviewed SQL | 83% confidence | Saved answer

```
SELECT
  e.department,
  COUNT(e.employee_id) AS headcount
FROM employees e
GROUP BY e.department
ORDER BY headcount DESC;
```

List Employees | Show Employees with Full Name | Count Employees by Department | Show Employees from this month

Ask for SQL in natural language

Speak or type a business question

SQL (standard) | Voice | Generate SQL

QUERY REVIEW

Query review

Reviewed

A concise view of the checks behind this response.

CONFIDENCE

83%

REVIEW SCORECARD

Plannin g 95%

Validati on 100%

Data fit 88%

7-STEP REVIEW FLOW

STAGE 01
Understand the request
Classified this as Count.

STAGE 02
Resolve business terms
Resolved business terms for Employees.

STAGE 03
Match relevant data
Matched Employees.

STAGE 04
Plan the query
Prepared a read-only plan with the matched business data.

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SQL Repair

Correct read-only SQL, identify syntax problems, and understand reported database connection errors.

Check a SQL statement

Paste a statement and, if available, the database error that was returned.

SQL (standard)

SQL statement

```
SELEC employee_id FROM employees
```

Database error (optional)

```
For example: ORA-12541: TNS:no listener, connection refused, timeout, or authentication error
```

The checker does not open a database connection or execute SQL.

Check and correct

Repair result

Syntax, safety, and reported connection findings.

Ready to review

VALIDATION

Syntax parsed successfully.

CORRECTED STATEMENT

Copy

```
SELECT employee_id FROM employees;
```

STATEMENT FINDINGS 2

warning **Common syntax issue corrected**

Corrected the SELECT keyword.

info **Statement terminator added**

Added a semicolon to complete the statement.

CONNECTION DIAGNOSTICS

Add the database error message above to diagnose listener, port, DNS, timeout, or sign-in problems.

Connection guidance stays safe

SQL Copilot

From business context to reviewed SQL

A schema-aware workflow for turning questions into safe, explainable SQL with visible review checkpoints.

- **Schema aware** Business vocabulary, table mapping, relationships, and join paths
- **Reviewable** Seven workflow stages, validation, confidence, and repair guidance
- **Deployed** Cloudflare frontend, Oracle API, and NVIDIA structured generation

Try it live: sql-copilot.pages.dev

github.com/shasanksingh/sql-copilot